CS 385, Homework 4: Decrease/Divide and Conquer

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Point values are assigned for each question. Points earned: ____ / 100

1. Consider the algorithm on page 148 in the textbook for Binary Reflected Gray Codes. What change(s) would you make so that it generates the binary numbers **in order** for a given length `n`? Your algorithm must be recursive and keep the same structure as the one in the textbook. Describe only the change(s). (10 points)

   Most algorithms BRGC(n) would stay the same. However, in the else statement, instead of copying list one to list two in reverse order, we would copy it in the same order. "copy list L1 to list L2 in reversed order" would be "copy list L1 to list L2 in **the same order**"

2. Show the steps to multiply 72 x 93 with Russian peasant multiplication, as seen in Figure 4.11b on page 154 in the textbook. (10 points)

| | |
|---|---|
| 72 | 93 |
| 36 | 186 |
| 18 | 372 |
| **9** | **744** |
| 4 | 1488 |
| 2 | 2976 |
| **1** | **5952** |

744+5952= 6696
**72 x 93=6696**

3. Suppose you use the LomutoPartition() function on page 159 in the textbook in your implementation of Quicksort. (10 points, 5 points each)
   a) Describe the types of input that cause Quicksort to perform its worst-case running time.

Quicksort would perform its worst-case running time in the LomutoPartition() function when the sequence of numbers is sorted in reverse order.

   b) What is that worst-case running time?

When this happens, the worst-case running time would be $\Theta(n^2)$.

4. Compute 2205 x 1132 by applying the divide-and-conquer Karatsuba algorithm outlined in the text. Repeat the process until the numbers being multiplied are each 1 digit. For each multiplication, show the values of $c_2$, $c_1$, and $c_0$. Do not skip steps. (10 points)

4. $2205 \cdot 1132 = 2496060$

$$\begin{array}{r} 2420000 \\ 75900 \\ +\quad 160 \\ \hline 2496060 \end{array}$$

$a1 = 22 \quad b1 = 11 \quad c2 = a1 \cdot b1 = 22 \cdot 11 = 242$

$a0 = 5 \quad b0 = 32 \quad c0 = a0 \cdot b0 = 5 \cdot 32 = 160$

$c1 = (a1 + a0) \cdot (b1 + b0) - (c2 + c0) = (22+5) \cdot (11+32) - (242+160)$

$= (27) \cdot (43) - 402 = 1161 - 402 = 759$

$22 \cdot 11 = 242$

$$\begin{array}{r} 200 \\ 40 \\ +\quad 2 \\ \hline 242 \end{array}$$

$a1 = 2 \quad b1 = 1 \quad c2 = a1 \cdot b1 = 2 \cdot 1 = 2$

$a0 = 2 \quad b0 = 1 \quad c0 = a0 \cdot b0 = 2 \cdot 1 = 2$

$c1 = (a1 + a0) \cdot (b1 + b0) - (c2 + c0) = (2+2) \cdot (1+1) - (2+2)$

$= 4$

$5 \cdot 32$

$$\begin{array}{r} 000 \\ 150 \\ +\quad 10 \\ \hline 160 \end{array}$$

$a1 = 0 \quad b1 = 3 \quad c2 = a1 \cdot b1 = 0 \cdot 3 = 0$

$a0 = 5 \quad b0 = 2 \quad c0 = a0 \cdot b0 = 5 \cdot 2 = 10$

$c1 = (a1 + a0) \cdot (b1 + b0) - (c2 + c0) = (0+5) \cdot (3+2) - (0+10)$

$= 5 \cdot 5 - 10 = 15$

$27 \cdot 43$

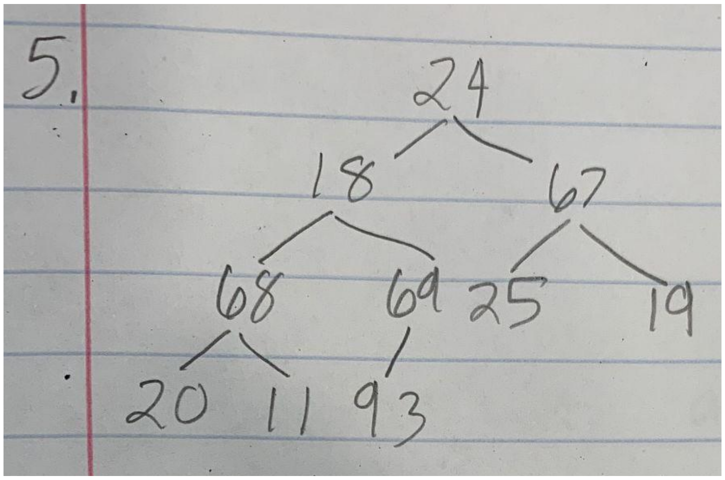
$$\begin{array}{r} 800 \\ 340 \\ +\quad 21 \\ \hline 1161 \end{array}$$

$a1 = 2 \quad b1 = 4 \quad c2 = a1 \cdot b1 = 2 \cdot 4 = 8$

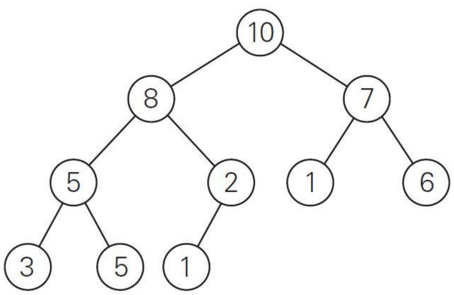
$a0 = 7 \quad b0 = 3 \quad c0 = a0 \cdot b0 = 7 \cdot 3 = 21$

$c1 = (2+7) \cdot (4+3) - (8+21) = 9 \cdot 7 - 29 = 63 - 29 = 34$

5. Draw the binary search tree after inserting the following keys: 24 18 67 68 69 25 19 20 11 93 (10 points)




6. Consider the following binary tree. (16 points, 2 points each)




a) Traverse the tree preorder.

10, 8, 5, 3, 5, 2, 1, 7, 1, 6

b) Traverse the tree inorder.

3, 5, 5, 8, 1, 2, 10, 1, 7, 6

c) Traverse the tree postorder.

3, 5, 5, 1, 2, 8, 1, 6, 7, 10

d) How many internal nodes are there?

5 internal nodes: 10, 8, 7, 5, 2

e) How many leaves are there?

5 leaves: 3, 5, 1, 1, 6

f) What is the maximum width of the tree?

The maximum width of this tree is 4 (composed of 5, 2, 1, 6)

g) What is the height of the tree?

The height of the tree is 3.

h) What is the diameter of the tree?

The diameter of the tree is 6.

7. Use the Master Theorem to give tight asymptotic bounds for the following recurrences. (25 points, 5 points each)



7\. a. $T(n) = 2T(n/4) + 1$ $\quad a=2 \quad b^k = 4^0 = 1 \quad 2>1$

$a > b^k$

$T(n) \in \Theta(n^{\log_4 2})$

b. $T(n) = 2T(n/4) + \sqrt{n}$ $\quad a=2 \quad b^k = 4^{1/2} = 2 \quad 2=2$

$a = b^k$

$T(n) \in \Theta(\sqrt{n} \cdot \log n)$

c. $T(n) = 2T(n/4) + n$ $\quad a=2 \quad b^k = 4^1 = 4 \quad 2<4$

$a < b^k$

$T(n) \in \Theta(n)$

d. $T(n) = 2T(n/4) + n^2$ $\quad a=2 \quad b^k = 4^2 = 16 \quad 2<16$

$a < b^k$

$T(n) \in \Theta(n^2)$

e. $T(n) = 2T(n/4) + n^3$ $\quad a=2 \quad b^k = 4^3 = 64 \quad 2<64$

$a < b^k$

$T(n) \in \Theta(n^3)$

8. Consider the following function. (9 points)

```
int function(int n) {
    if(n <= 1) {
        return 0;
    }
    int temp = 0;
```

```
        for(int i = 1; i <= 6; i++) {
            temp += function(n / 3);
        }
        for(int i = 1; i <= n; i++) {
            for(int j = 1; j * j <= n; j++) {
                temp++;
            }
        }
        return temp;
    }
```

a) Write an expression for the runtime $T(n)$ for the function. (4 points)

8. a. $T(n) = 6T(n/3) + n\sqrt{n}$

$T(n) = 6T(n/3) + n^{3/2}$

b) Use the Master Theorem to give a tight asymptotic bound. Simplify your answer as much as possible. (5 points)

b. $T(n) = 6T(n/3) + n^{3/2}$ $a = 6$ $b^k = 3^{3/2} = \sqrt{27} = 3\sqrt{3}$

$6 > 3\sqrt{3}$ so $a > b^k$

$T(n) \in \Theta(n^{\log_3 6})$

I pledge my honor that I have abided by the Stevens Honor System. Octavio Morales